

The instrument is not the bottleneck

What BEOPS is for, argued against the literature rather than asserted

Status: analysis. Every empirical claim below is either measured in this project's own registers or taken from a named source with a link. Date: 2026-09-09. Authors: prof. dr Darinka Golubović Matić · doc. dr Semir Poturak — a scientific paper prepared for the conference "Creating sustainable commUNiTy" at University Union – Nikola Tesla. Not a product, not a service, and not the institution speaking.

1. The claim

A city does not become knowable by acquiring more instruments. Belgrade already has a state air quality network, meteorological and hydrological stations, an airport that reports the weather every half hour, a distribution utility that publishes tomorrow's outages street by street, a water utility that announces its own failures, a parking operator that counts empty spaces, and a citizen sensor network. What a resident still cannot do is answer a simple question — *is the air bad in my part of the city right now, how sure is anyone, and when was that actually measured* — and take a decision on the answer.

The gap is therefore not a **sensing** gap. It is an **epistemic** gap: the published record is scattered, its times mean different things, its provenance is not stated, its absences are invisible, and its terms of reuse are unrecorded. That is the gap this work addresses, and the reason it is a records project before it is a display project.

2. The evidence that instruments are not the bottleneck

The largest deployment says so itself. The Newcastle Urban Observatory runs, by its own account, more than 3,600 sensor streams across 65 variables, ingesting on the order of 10,000 observations a minute and having collected over ten billion city observations. In the paper describing it, the team writes plainly: *"Few people want data, what they really want is analysis and insight,"* and reports that this analysis remains "manual, time consuming, and expensive." They also describe the parts that rarely reach a dashboard — sensors developing faults and reporting excessive values, indicative air sensors permitted to sit within 50% of a reference instrument, and a maintenance burden that becomes substantial past a certain number of units (James et al., 2022, *Frontiers in Sustainable Cities* 4, 767942, CC BY).

The lesson is not that Newcastle failed. It is that a city with three orders of magnitude more instruments than this project reads still identifies the same bottleneck: the step from observation to usable knowledge.

What the presentation layer routinely omits. Kitchin and McArdle's audit of city dashboards names six issues; two of them are exactly the ones a resident runs into. On *veracity and validity*, they find that "data quality metadata is rarely provided," while dashboards "demand trust without transparency." On *scope and access*, they report from their own Dublin audit that only **10 of 37** sustainability indicators were available at an appropriate scale — the rest were missing, wrongly scaled, or locked away (Kitchin & McArdle, *Urban data and city dashboards: six key issues*, Programmable City Working Paper 21; see also Kitchin, Lauriault & McArdle, 2015, *Regional Studies*, *Regional Science* 2(1), doi:10.1080/21681376.2014.983149).

The same audit, run for Belgrade, produces a number. This project's registry is that audit: 211 recorded access routes, of which 28 are actually polled, 14 publishers said no, and 26 remain undocumented. A reader can check every one of those states against stored bytes. The claim "the city publishes less than it appears to" stops being rhetoric at that point and becomes a figure someone can dispute.

A yardstick exists. ISO 37120 (2018; revised 2023) defines indicators for city services and quality of life, and ISO 37122 extends it for so-called smart cities. They give an external, neutral list against which a city's *answerable* questions can be counted rather than asserted. Mapping the Belgrade registry onto that list — how many ISO 37120 indicators can be answered today from open sources, at what latency, with what provenance — is the smallest genuinely original measurement this project can offer, and it is not yet done.

3. Where AI belongs in this argument, and where it does not

The tempting sentence is "AI turns the city's data into decisions." It should be resisted, because it walks directly into the strongest critique in this literature: Kitchin and McArdle's fifth issue, where dashboards carry an instrumental rationality that treats the city as a controllable system and can displace deliberative politics. A project that answers "we added AI" to that critique has not answered it.

The defensible claims are narrower and, we think, more interesting:

1. **AI lowers the cost of integration, which is where the labour actually is.** The Newcastle team's "manual, time consuming, and expensive" is a description of parsing, aligning and maintaining heterogeneous public sources. That is work models genuinely reduce.
2. **A model can narrate a record, but only behind a gate.** In this system the local models may use only numbers that appear in the digest built from the collected rows; they must cite the fact they are speaking about, inside the sentence; they may not state a forecast as a fact; and every utterance is validated before a reader sees it. Refusals are published as refusals.
3. **The risk is now named in the literature and it is exactly the risk this design fears.** Huang (2025) defines *geo-hallucination* as model output that is "spatially incoherent or fabricated, despite

appearing authoritative," distinguishes it from ordinary data error because it comes from the model's reasoning rather than the measurement, and calls it an epistemic and socio-political risk for planning, governance and equity. The recommendation is grounding, critical literacy and a renewed commitment to evidence-based approaches with human oversight (*Environment and Planning B: Urban Analytics and City Science*, doi:10.1177/23998083251409526).

So the honest formulation is: **the contribution is the gate and the record, not the model**. An AI that can say anything about a city is a liability; an AI that can only say what the record supports, and whose refusals are as visible as its sentences, is an instrument. The second is what is being built here, and it is testable — the refusals are counted on the public page.

4. What this project therefore claims, in one paragraph

Belgrade's problem is not too few instruments but too little defensible record. This work builds the missing intermediate object: every value carried with its source, its three times, its unit, its state among five (observed, untimed, estimated, forecast, unavailable), and the captured permission that allowed it to be read at all; absence recorded as a positive fact with its reason; corrections appended and never erased. On top of that record, and only on top of it, small local models comment in Serbian behind a validator that admits only what the record supports. The instruments were already there. What was missing was the bookkeeping that makes them usable for a decision — and the honesty about how much is still missing.

5. Honest verdict

- The argument in §2 is strong because it is not ours: the largest observatory in the comparison says the bottleneck is insight, and the standard critique of dashboards says the missing thing is quality metadata. Our contribution is to have built the metadata layer, for one city, in public.
 - The argument in §3 is where a hostile reviewer will push, and rightly. "The gate works" is currently supported by counted refusals over a few days on one machine, with three small local models. It is a demonstration, not an evaluation. An evaluation would need an adversarial test set of digests and utterances, and a false-accept rate. That is the next piece of work and it is not done.
 - The ISO 37120 mapping in §2 is proposed, not performed. Until it is, "the city publishes less than it appears to" rests on our own registry categories rather than an external standard.
 - Nothing here shows that better bookkeeping improves any actual decision. That claim would need a user study with people who take decisions, and this project has not run one. The most that is claimed is that a decision cannot be well made on a number whose age, origin and permission are unknown.
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Sources cited above

- James, P., Jonczyk, J., Smith, L., Harris, N., Komar, T., Bell, D., & Ranjan, R. (2022). Realizing Smart City Infrastructure at Scale, in the Wild: A Case Study. *Frontiers in Sustainable Cities*, 4, 767942. <https://doi.org/10.3389/frsc.2022.767942> (CC BY)
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- Kitchin, R., Lauriault, T. P., & McArdle, G. (2015). Knowing and governing cities through urban indicators, city benchmarking and real-time dashboards. *Regional Studies, Regional Science*, 2(1). <https://doi.org/10.1080/21681376.2014.983149>
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- ISO 37120:2018 (rev. 2023), Sustainable cities and communities — Indicators for city services and quality of life. <https://www.iso.org/standard/68498.html>